AI-Powered Sanskrit NLP Framework for Morphological and Semantic Analysis

IT Project Management (21MGH303P)

B.Tech – III Year | 6th Semester

(AI-driven framework for morphological analysis, semantic parsing, and structured knowledge extraction from classical Sanskrit texts based on Paninian grammar.)

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Problem Statement

- **Current Challenges in Sanskrit NLP**
  - Lack of integrated systems combining Paninian grammar with modern AI techniques
  - Existing tools are primarily:
    - Rule-based (limited scalability)
    - Or statistical (lack linguistic depth)
- **Technical Limitations**
  - Inefficient morphological analysis for complex Sanskrit structures
  - Absence of robust semantic parsing frameworks
  - Limited capability for structured knowledge extraction
- **Research & Practical Gaps**
  - Minimal support for large-scale corpus processing
  - Poor accessibility for researchers (lack of standardized APIs)
  - High dependency on manual linguistic analysis

Project Objectives

Primary Objectives

- Develop a scalable NLP framework for Sanskrit text analysis
- Ensure integration of Paninian grammar with AI techniques

Management Objectives

- Deliver project within:
- Budget: ₹33–41 Lakhs
- Timeline: 218 Days
- Maintain quality standards (≥92% accuracy target)

Operational Objectives

- Provide API-based access for researchers
- Enable large-scale corpus processing
- Ensure ease of use and adoption

Strategic Goal

- Establish a sustainable and extensible platform for academic and research use

Requirements & Scope Overview

Requirement Approach

- **Requirements gathered via:**
  - Stakeholder workshops
  - Document analysis
  - Institutional survey

Key Requirement Categories

- Functional: Core system capabilities (analysis, API access, processing)
- Non-Functional: Performance, accuracy, scalability, reliability

Scope Definition

In Scope

- NLP framework development
- API deployment and documentation
- Cloud-based processing

Out of Scope

- Mobile application
- Real-time streaming systems
- OCR / speech processing

Priority Management

- Requirements prioritized using MoSCoW method
- Focus on critical deliverables for initial release

Stakeholder Management

Key Stakeholders

- Sponsors (UGC / MHRD)
- Sanskrit Linguists
- NLP Engineers & Data Team
- Researchers & End Users
- Cloud Service Provider

Communication Strategy

- Weekly reviews (core team)
- Monthly reporting (sponsors)
- Feedback cycles (end users)

Management Approach

- Stakeholders classified based on Power vs Interest
- Engagement strategies:
  - High Power–High Interest → Close monitoring
  - Others → periodic communication

Outcome

- Improved decision-making
- Reduced project risks
- Better stakeholder alignment

Stakeholder Classification Framework

Stakeholders are mapped based on:

- Power (influence on project)
- Interest (level of involvement)

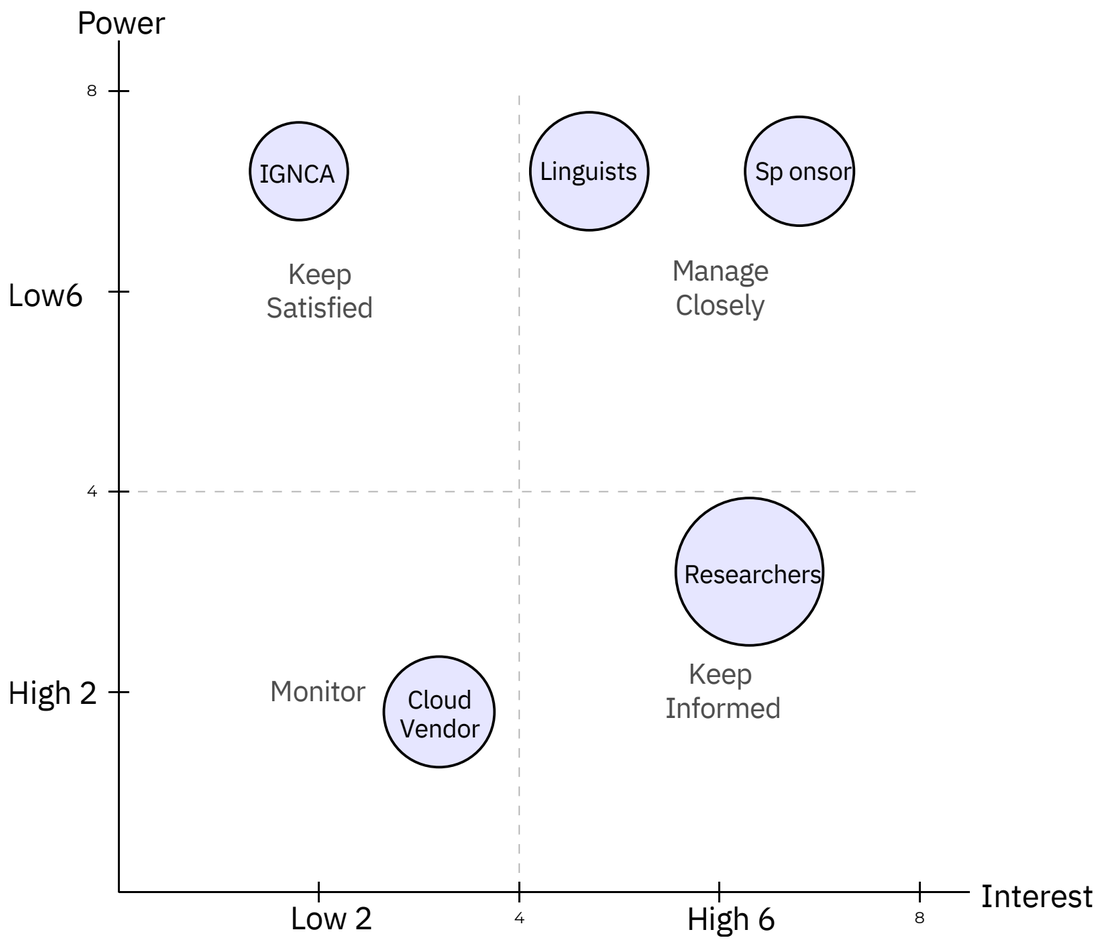
Management Strategy

- **High Power – High Interest:** Manage closely
- **High Power – Low Interest:** Keep satisfied
- **Low Power – High Interest:** Keep informed
- **Low Power – Low Interest:** Monitor

Project Insight

- **Critical stakeholders (Sponsors, Linguists) require continuous engagement**
- **End users require regular updates and feedback integration**

Stakeholder Power/Interest Grid



Requirement Prioritization (MoSCoW)

Purpose

- Ensure focused **delivery within time** and budget constraints
- Prioritize requirements based on **business value and criticality**

Categories

- **Must Have:** Essential for project success
- **Should Have:** Important but not critical
- **Could Have:** Value-adding enhancements
- **Won't Have:** Out of current scope

Management Insight

- Enables **controlled scope and prevents scope creep**
- Supports **phased delivery** strategy

Requirement Prioritization (MoSCoW)

Must Have - Morphological Analyzer - Semantic Parser - Core NLP Engine	Should Have - REST API - Batch Processing - Knowledge Extraction
Could Have - UI Visualization - Bilingual Interface	Won’t Have -OCR - Speech Processing - Translation

Work Breakdown Structure (WBS)

Purpose

- Decompose the project into **manageable** components
- Improve **planning, scheduling, and resource allocation**

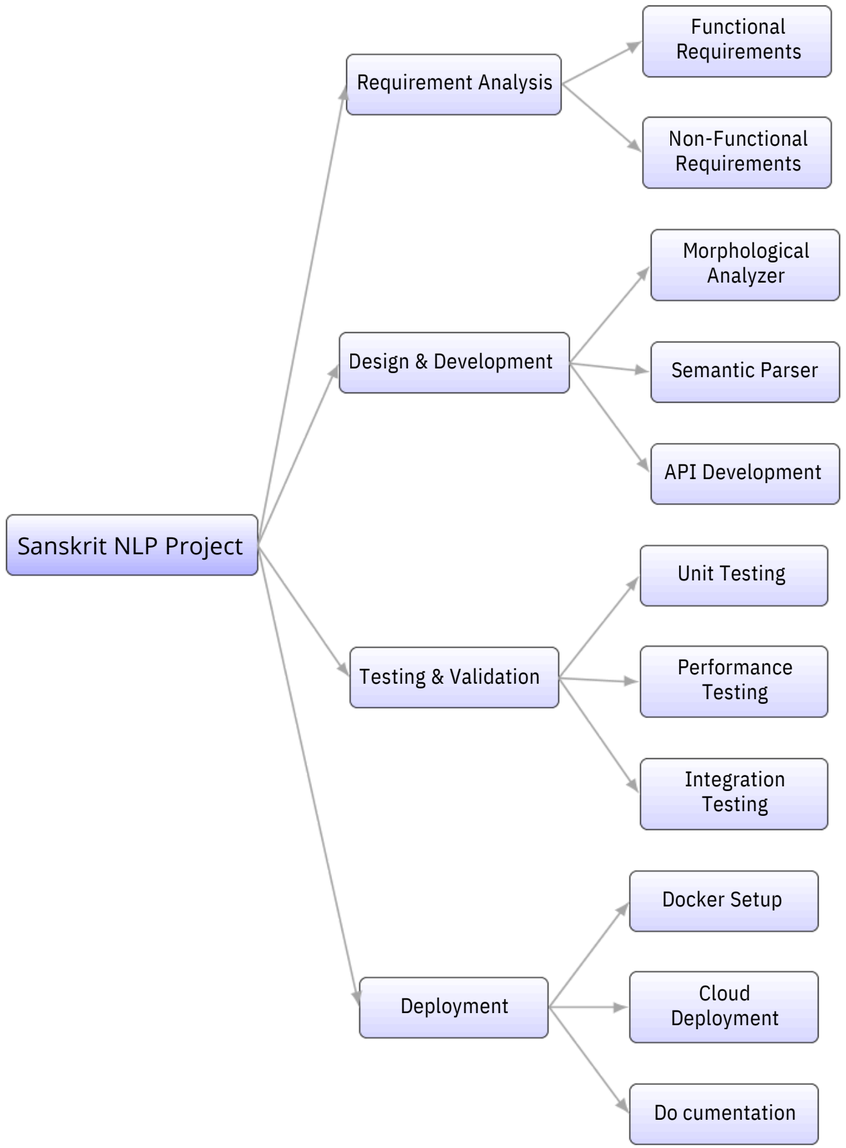
Structure

- **Level 1:** Project
- **Level 2:** Major phases **(Analysis, Development, Testing, Deployment)**
- **Level 3:** Subtasks

Management Insight

- Enables **clear task ownership**
- Supports accurate **time and cost estimation**
- Forms the basis for **PERT, CPM, and scheduling**

Work Breakdown Structure (WBS)



Critical Path Method (CPM) Analysis

Purpose

- Identify the **longest path of dependent activities**
- Determine **minimum project completion time**

Key Insight

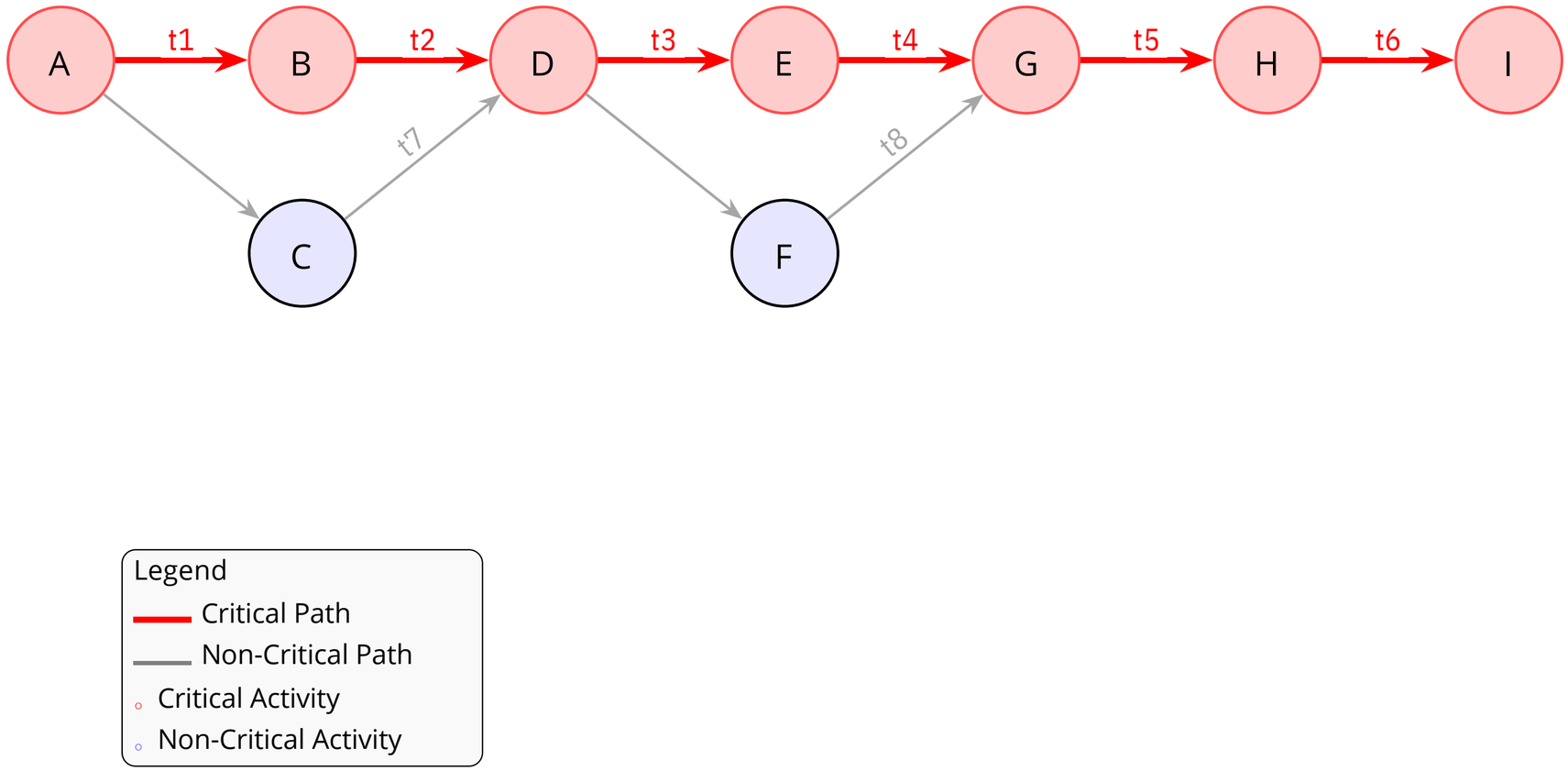
- **Critical Path: A → B → D → E → G → H → I**
- **Total Duration: 218 Days**

Management Implications

- Activities on critical path have zero slack
- Any delay directly impacts project timeline
- Enables focused monitoring and control

Critical Path Method (CPM) Analysis

Critical Path Method – Project Network



Project Schedule

Purpose

- Visualize **project timeline and task sequencing**
- Track **start and end dates of activities**

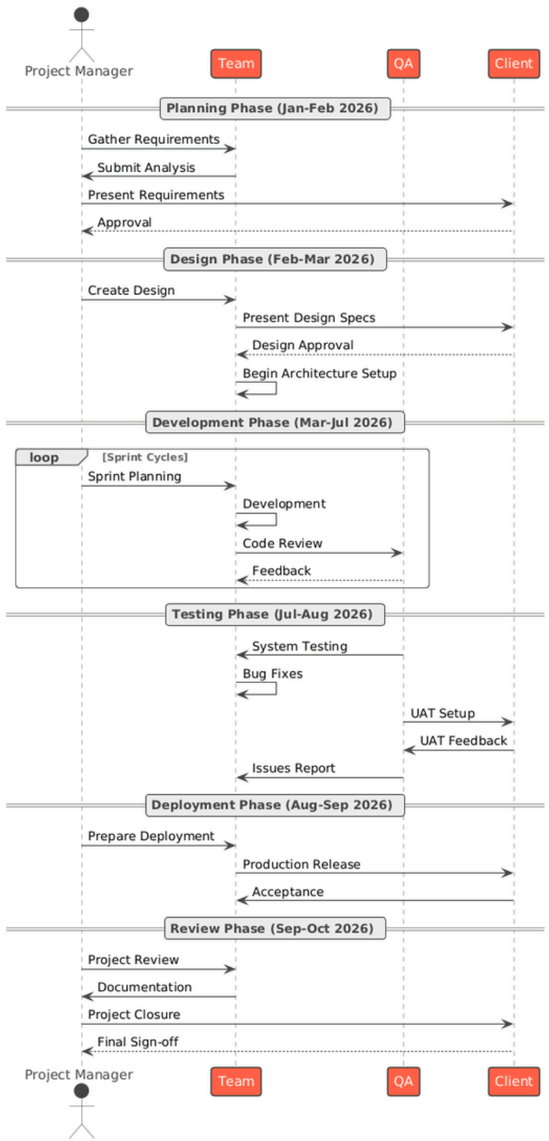
Schedule Overview

- **Total Duration: 218 Days (~10.5 Months)**
- **Phases:**
  - **Requirement Analysis**
  - **Design & Development**
  - **Testing**
  - **Deployment**

Management Insight

- Enables progress tracking against plan
- Helps identify overlapping tasks and delays
- Supports resource planning and coordination

Project Schedule(Gantt Chart)



Project Performance Monitoring

Objective

- **Track project performance across:**
  - **Cost**
  - **Schedule**
  - **Scope**

Monitoring Approach

- Use of **Key Performance Indicators (KPIs)**
- Regular **status reviews and reporting**
- Comparison of **planned vs actual performance**

Control Mechanisms

- **Variance analysis**
- **Corrective action planning**
- **Continuous performance evaluation**

Management Insight

- Early identification of **delays and cost overruns**
- Enables **data-driven** decision making
- Ensures project remains aligned with **baseline plan**

Earned Value Analysis (EVA)

Purpose

- Evaluate project **performance** using **cost and schedule metrics**
- Compare **planned vs actual vs earned** progress

Key Metrics

- **Planned Value (PV):** Budgeted work
- **Earned Value (EV):** Work completed
- **Actual Cost (AC):** Cost incurred

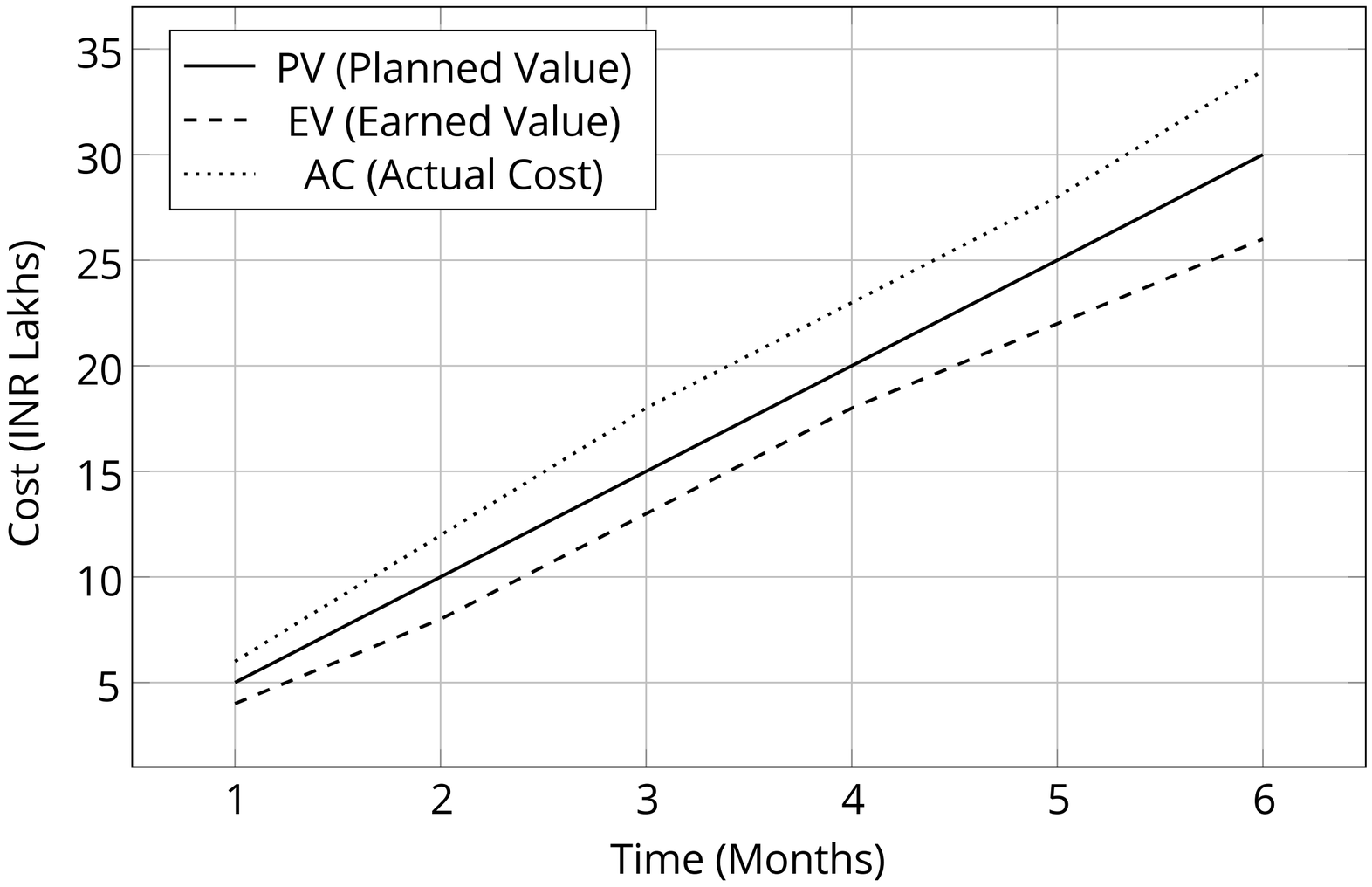
Performance Indicators

- **CPI (Cost Performance Index)**
- **SPI (Schedule Performance Index)**

Management Insight

- **EV < PV → Schedule delay**
- **EV < AC → Cost overrun**
- Supports forecasting and corrective actions

Earned Value Analysis (EVA)



Project Risk Management

Objective

- **Identify, assess, and control** project risks proactively

Risk Management Process

- Risk Identification
- Risk Analysis (Probability & Impact)
- Risk Response Planning
- Risk Monitoring

Key Risks Identified

- Complexity of linguistic rules
- Resource constraints (expert availability)
- Data annotation delays

Management Strategies

- **Risk mitigation** and **contingency** planning
- Allocation of risk reserves
- Continuous monitoring through **RMMM plan**

Outcome

- Reduced uncertainty
- Improved project stability and predictability

Risk Assessment Matrix

Project Risks

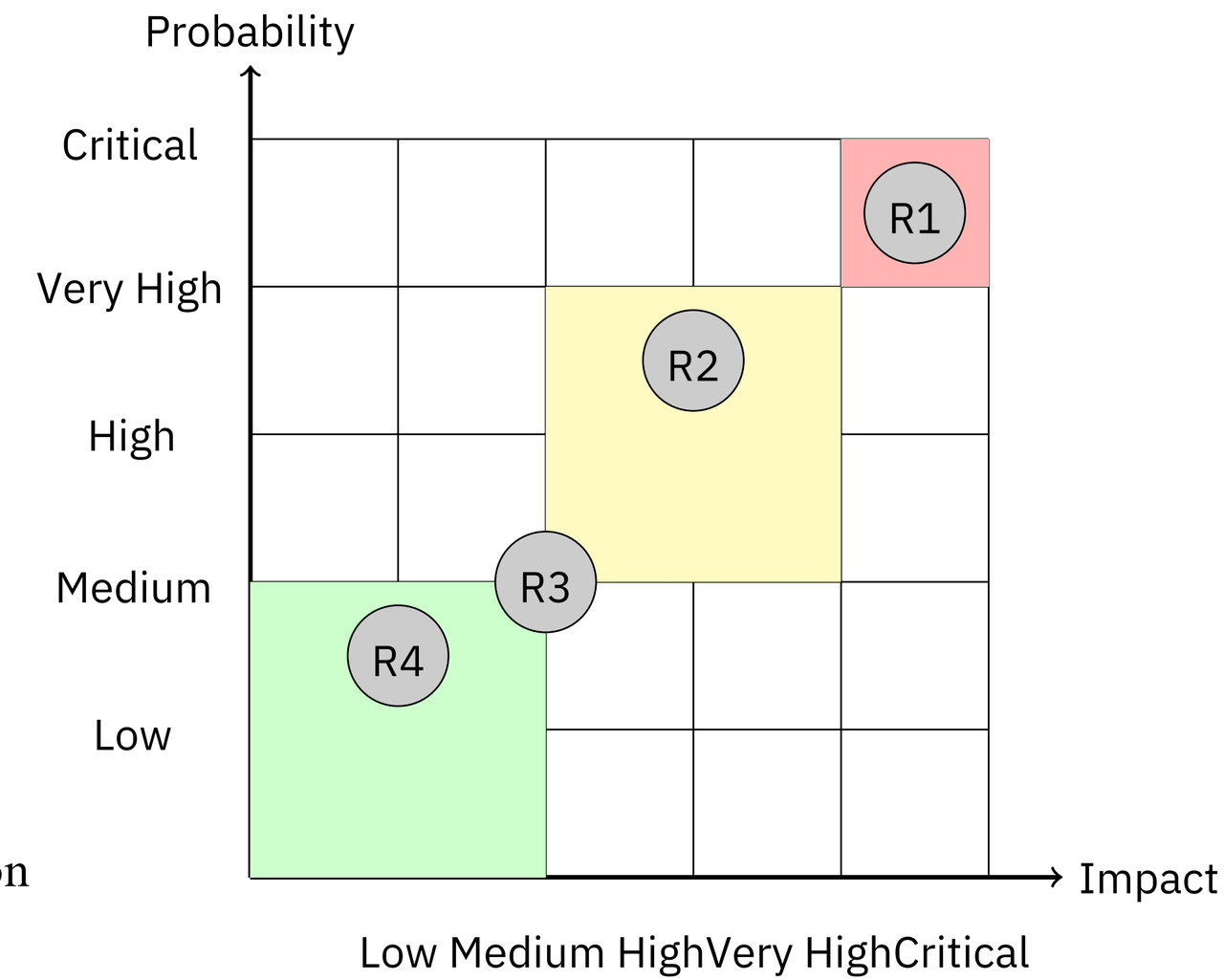
- **R1:** Paninian grammar complexity
- **R2:** Limited expert availability
- **R3:** Data annotation delays
- **R4:** Minor operational inefficiencies

Risk Positioning

- **R1 → Critical Zone**
  - High probability + very high impact
- **R2 → High Risk Zone**
  - Moderate–high probability and impact
- **R3 → Medium Risk Zone**
  - Moderate impact, manageable
- **R4 → Low Risk Zone**
  - Low probability and impact

Management Insight

- Focus on **R1** and **R2** for immediate mitigation
- Monitor **R3**, accept/manage **R4**



Agile Approach

Rationale for Agile Adoption

- **Evolving requirements** due to **complex** Sanskrit grammar rules
- Need for **iterative validation** with linguists
- **Continuous feedback** from research stakeholders

Framework Used

- Scrum **(2-week sprints)** for **development & testing**
- Integrated with **XP** practices for quality

Sprint Structure (Project Context)

- **Sprint Duration: 2 Weeks**
- **Activities:**
  - Requirement refinement (with linguists)
  - Feature development (NLP modules)
  - Testing & validation
  - Review with stakeholders

Scrum Framework

Workflow in This Project

- **Product Backlog: NLP features (morphology, parsing, API)**
- **Sprint Planning:** Prioritization based on **MoSCoW**
- **Sprint Backlog:** Selected tasks for each sprint
- **Sprint (2 Weeks):** Development + testing cycle

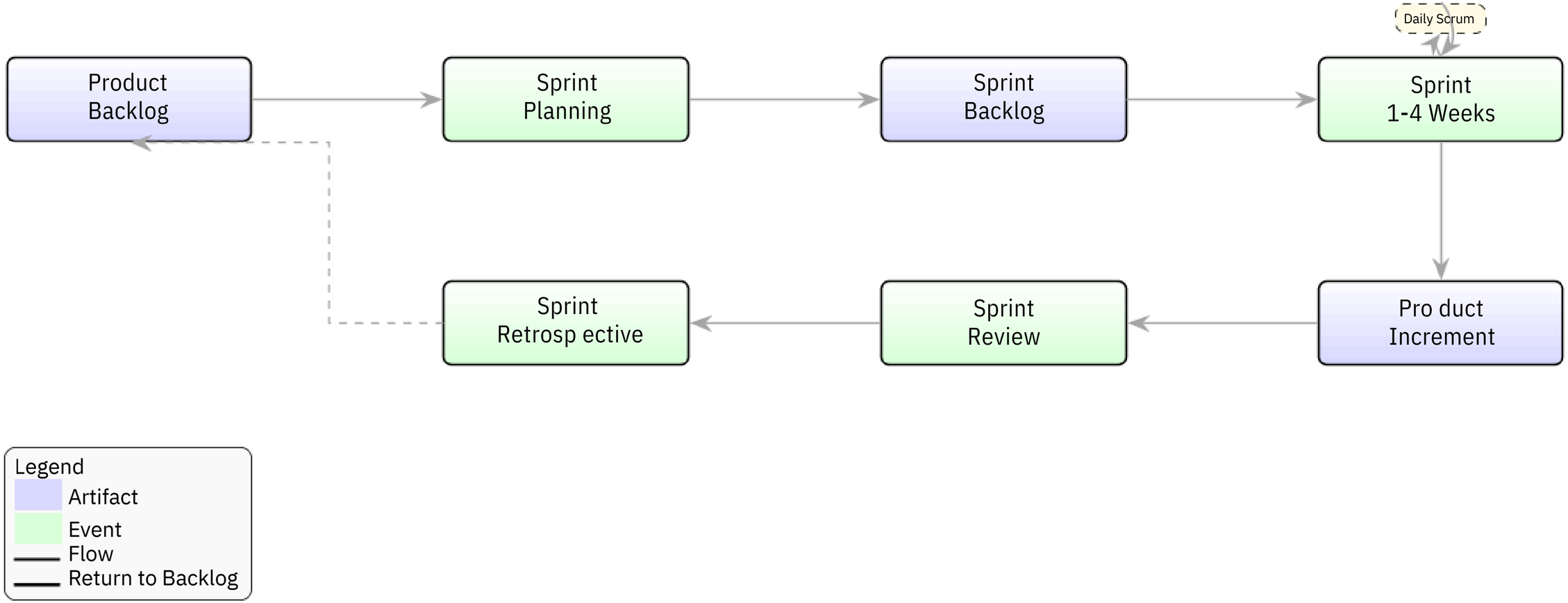
Execution Cycle

- **Daily Scrum → progress tracking**
- **Sprint Review → stakeholder validation (linguists)**
- **Sprint Retrospective → process improvement**
- **Output → Incremental feature delivery**

Project-Specific Insight

- Ensures continuous validation of linguistic accuracy
- Supports iterative delivery of NLP modules
- Improves stakeholder collaboration

Scrum Framework



Sprint Progress Tracking

Sprint Context

- **Sprint Duration: 10 Days (2 Weeks)**
- **Initial Workload: 50 Story Points**
- **Scope: NLP module development + testing**

Performance Observation

- Actual progress remains above ideal line
- Indicates slower completion rate

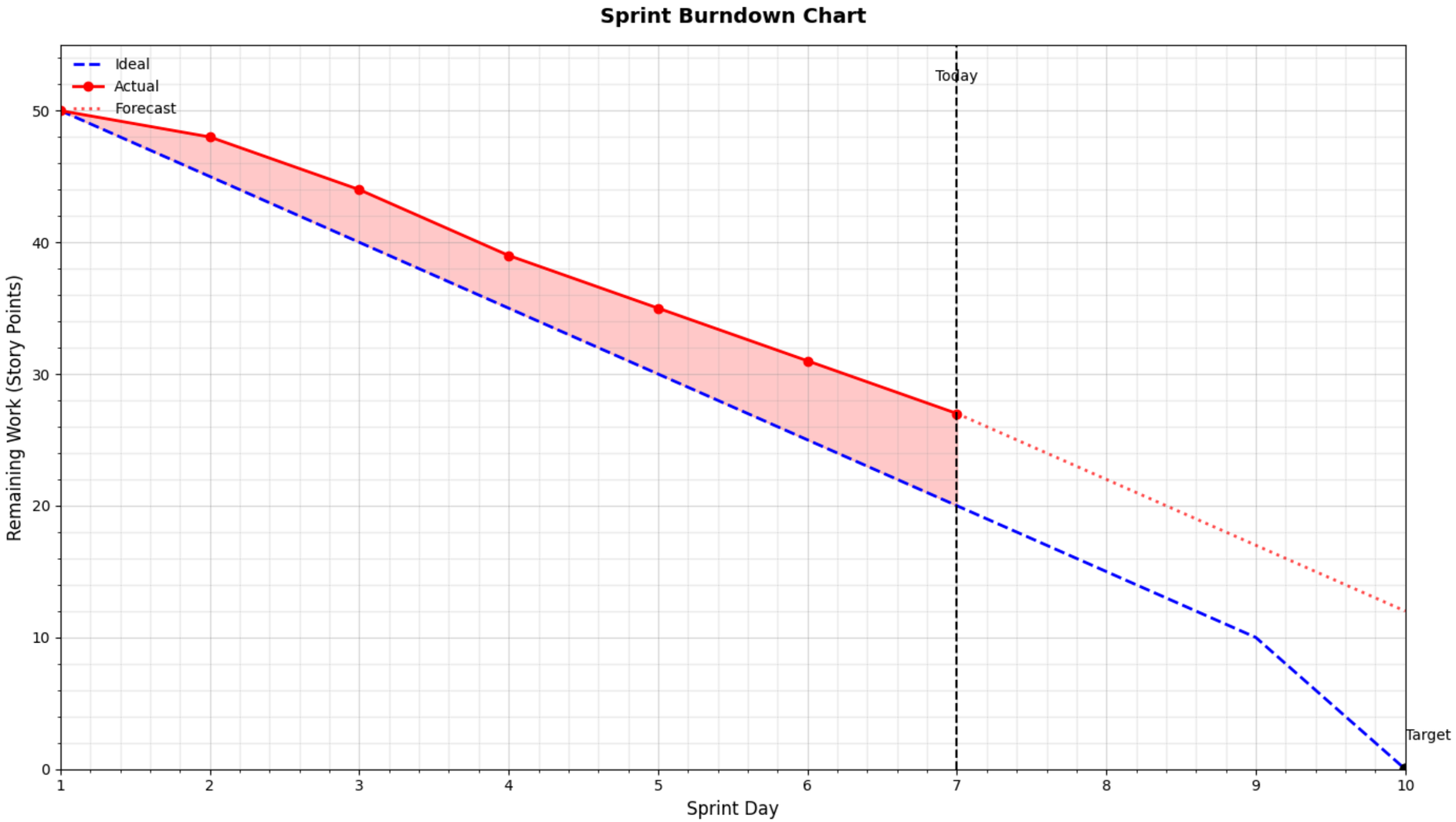
Forecast Insight

- Remaining work at sprint end → ~12 story points
- Risk of spillover to next sprint

Management Action

- Reallocate resources for critical tasks
- Improve task breakdown in next sprint
- Increase monitoring through daily scrums

Sprint Progress Tracking



DevOps & CI/CD Implementation

Objective

- Ensure reliable deployment of NLP framework
- Enable continuous integration of linguistic updates

Pipeline in This Project

- **Code Commit:** NLP modules (morphology, parser, API) via **GitHub**
- **Build: Integration of rule-based + ML components**
- **Test:**
  - **Morphological accuracy validation**
  - **API response testing (< 2 sec target)**
- **Package: Docker container** with NLP environment
- **Deploy: Cloud deployment** for research access

Project Insight

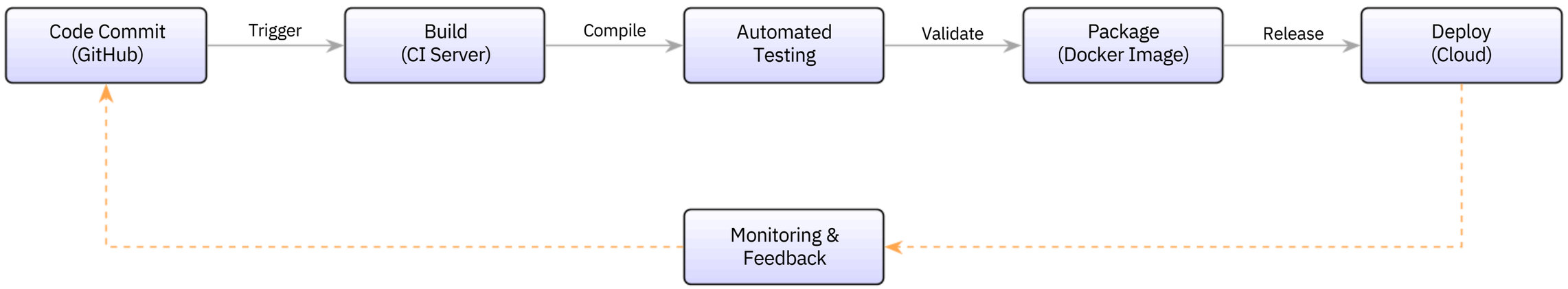
- Supports frequent updates of grammar rules (Panini sutras)
- Ensures consistent execution across environments
- Enables rapid bug fixes and feature releases

Management Impact

- **Reduces** deployment risk
- Improves delivery speed
- Enhances system reliability for stakeholders

DevOps & CI/CD Implementation

CI/CD Pipeline



Project Results & Outcomes

Project Performance

- **Total Cost: ₹33–41 Lakhs**
- **Project Duration: 218 Days (~10.5 Months)**
- **Payback Period: 3–4 Years**

Financial Evaluation

- **Net Present Value (NPV): Positive (~₹12.6 Lakhs)**
- **Indicates financial feasibility and long-term viability**

Operational Outcomes

- Achieved target accuracy (**~92%**) for analysis
- **Successful development of:**
  - **Morphological analyzer**
  - **Semantic parser**
  - **API-based system**

Management Outcome

- Delivered within planned scope, cost, and schedule
- **Effective use of:**
  - **Agile (Scrum)**
  - **Risk management**
  - **CI/CD practices**

Overall Impact

- Scalable platform for academic and research use
- Supports digital preservation and linguistic analysis

Conclusion

Project Summary

- Successfully planned and managed an AI-driven Sanskrit NLP framework
- Integrated traditional linguistic rules with modern technologies

Key Achievements

- Effective control of time, cost, and scope
- **Implementation of:**
  - **Structured planning (WBS, CPM)**
  - **Monitoring (EVA, Risk Management)**
  - **Agile & DevOps practices**

Managerial Insight

- Importance of **stakeholder collaboration (linguists + engineers)**
- Need for **adaptive planning in complex domains**
- Value of **continuous monitoring and iterative delivery**

Thank You!